

Final Report team 1

Text analytics application to enhance

Decathlon's Brand Image

Case Studies for DS & MA

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agreement
only a few pages
are available

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1. Final report

Please apply the following tentative structure in your report:

- Management summary
- Introduction with problem formulation
- Substantive literature review
- Data description
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- Conclusion
 - Data-driven actionable recommendations for the company
 - Advice for other researchers to better answer similar research problems
- Reflection on the project and lessons learned
 - How did you perform, as a team, on the six talents?
 - What can be improved and what are you most proud of?
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Management summary

For an e-commerce company like Decathlon, maintaining a strong brand image is crucial. It reflects the brand's values, product quality, and commitment to customer satisfaction, playing a pivotal role in attracting and retaining customers in the competitive online retail market. Decathlon assesses its brand image through customer reviews and ratings, with positive experiences leading to trust and credibility. Conversely, negative reviews can impact the brand negatively. Both positive and negative feedback serves as valuable insights for Decathlon to enhance customer experience and, consequently, its brand image.

The main goal of this report is to find valuable insights within customer reviews that can greatly improve both brand image and customer experience through identifying the strengths and weaknesses of their products. These strengths and weaknesses will be analysed to provide insights that Decathlon can use to improve its customers' experience and brand image.

The reviews were analysed through conducting sentiment analysis to extract the feelings of customers within reviews. The feedback found in the reviews was divided through topic analytics into 11 categories. The primary concerns extracted from the negative feedback revealed challenges with product storage and usage, comfort and wearability of products, and the effectiveness of the delivery service. On the other hand, the analysis identified key strengths in product utility, delivery efficiency, and the accuracy of product size and appearance descriptions, which resonate well with the customer base.

Decathlon can enhance the customer experience by improving product availability information, ensuring reliable delivery timelines, conducting quality checks, and providing accurate product descriptions to reduce returns. Clear communication on pricing, emphasising the quality of materials, can help customers better understand and accept product costs. Moreover, aligning product offerings with seasonal demand and material preferences can further improve the overall customer experience.

However, the study has some limitations because of the amount of data and the level of detail in the analysis. It suggests that future research should look more closely at what customers say about Decathlon on a product level and who the customers are based on demographic data. This could help Decathlon know their customers better and come up with more specific ways to improve the shopping experience online.

Results

Sentiment analysis

The analysis begins with the application of sentiment analysis on the sentences derived from the feedback of online channels. The calculated value is stored and assigned in the dataset as `sentiment_score`, as it will be used to divide the dataset for the upcoming analysis. From the results it is observed that 62.5% of the sentences were considered expressing a positive feeling, while the rest 37.5% includes the sentences with the negative sentiment. The higher number of positive sentiment sentences aligns with the findings that most of the reviews are rated from 4 to 5 stars (Figure 4 in the appendix), which implies a favourable opinion. This pattern is also consistent with the usual J-shaped distribution of notes seen in customer review datasets, where extreme satisfaction is more commonly observed. Notably, the volume of feedback from online purchases far exceeds that of offline transactions, indicating the higher proportion of online sales in the merged dataset. The data therefore supports the purpose of the study to focus on the online channel, since it has a greater number of consumer insights and a bigger transaction volume, providing useful views for customer satisfaction analysis and strategic company improvement.

Topic analysis

The sentiment scores resulting from the sentiment analysis are used to divide the available dataset into two subsets (positive/negative), in order to proceed with the topic analysis in each one. In the plots below it can be observed the different metrics previously discussed that helped define the optimal number of topics for each category.

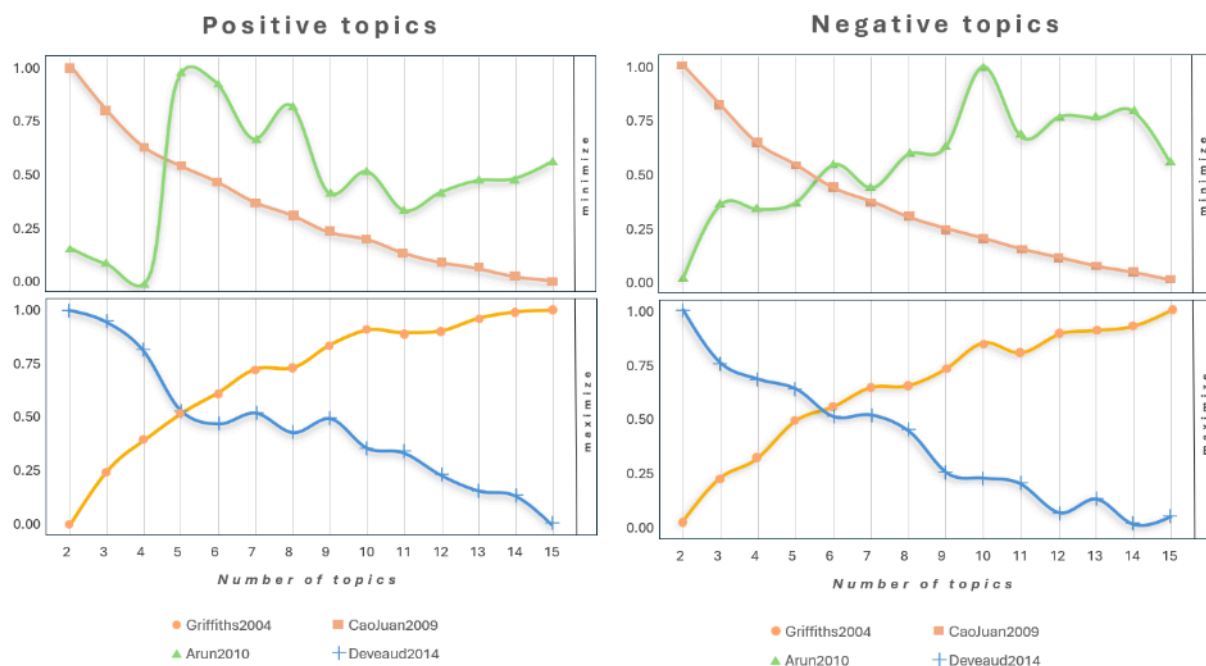
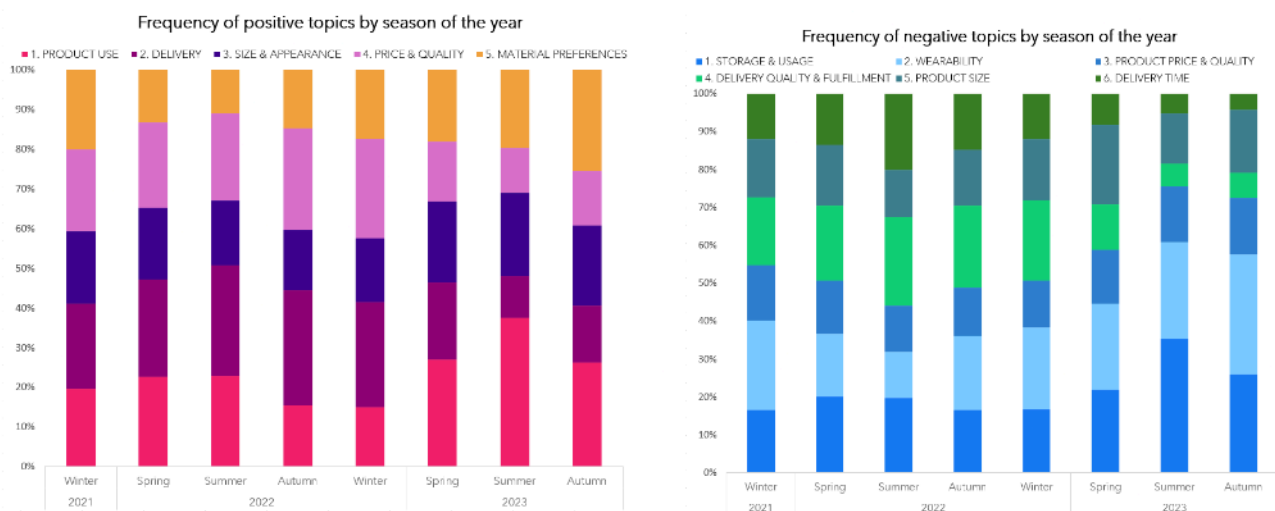


Figure 1. Plot metrics defining optimal number of topics. Left plot: optimal number of positive topics. Right plot: optimal number of negative topics

emphasising high-quality materials and affordable pricing for light clothing. These subtopics showcase customers' priorities for comfort, functionality, and quality in Decathlon products.

The positive theme "Delivery" reveals insights into customers' experiences with Decathlon's delivery services. Subtopics include "Customer Service" highlighting satisfaction with helpful staff; "Purchase in the Shop" focusing on in-store experiences; "Product Satisfaction and Appreciation" emphasising quality and functionality; "Efficient Ordering and Return" addressing ease of the process; "Timely Deliveries" discussing speed and accuracy; and "Day-to-Day Delivery Experiences" exploring package receipt and overall satisfaction. These subtopics offer valuable insights into customer satisfaction and preferences related to Decathlon's delivery services.

Seasonality



Conclusion

Recommendations

The detailed insights derived from the deeper topic analysis on the most negatively influential topics provide a foundation for recommending solutions to improve the online shopping experience. Highlighting the issues associated with the delivery process, which affects the rating of the reviews the most, implementing effective actions is crucial for improving the perception of the company. Constant updates of the stock availability in both warehouses and physical stores will lead customers to more informed purchasing decisions. Additionally, featuring likely restock dates on the website and allowing customers to sign up for notifications when desired products become available are essential steps towards transparency. Tracking process is also a critical area for improvement. By sending more regular emails with updated information about the delivery time, especially in cases where a heavy delay has occurred, a continuous and comforting customer experience can be achieved. Regarding return complaints, a detailed and easy to understand step by step guide within the packaging for the returning process will provide a straightforward and more accessible